Electronic Supplementary Information

Recall, not verification, is the bottleneck when frontier LLMs elucidate molecular structures from real spectra

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This document contains the Supporting Information figures referenced in the main article. Data, predictions, and the code that regenerates every figure are released with the manuscript; see *Data and code availability* in the main text.

Fig. S1: Study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.

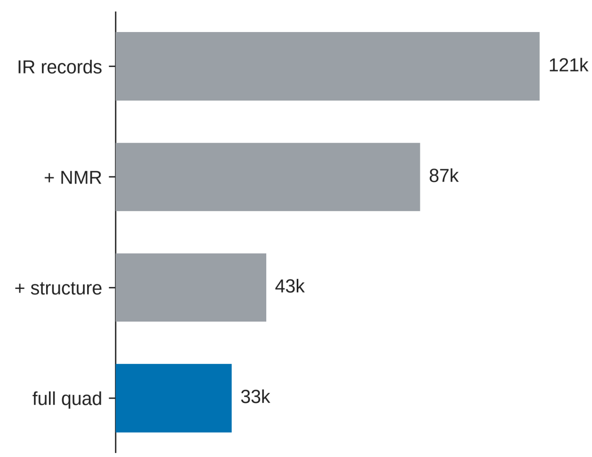

Fig. S2: IRexp composition: IR records, NMR-paired, structure-linked, and full IR+^{1}H+^{13}C+structure quadruples.

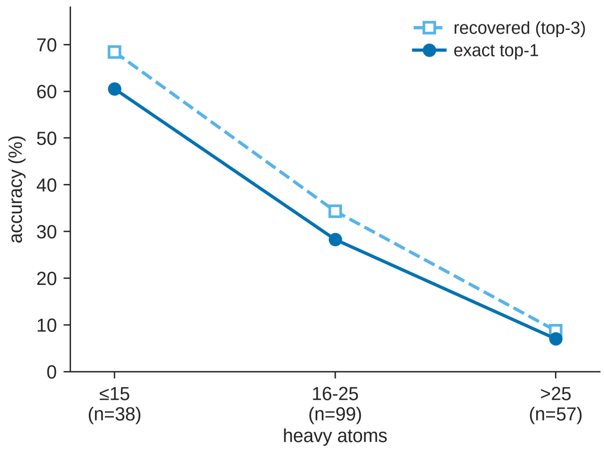

Fig. S3: Accuracy versus molecular size; the monotonic 60%→7% top-1 gradient with heavy-atom count.

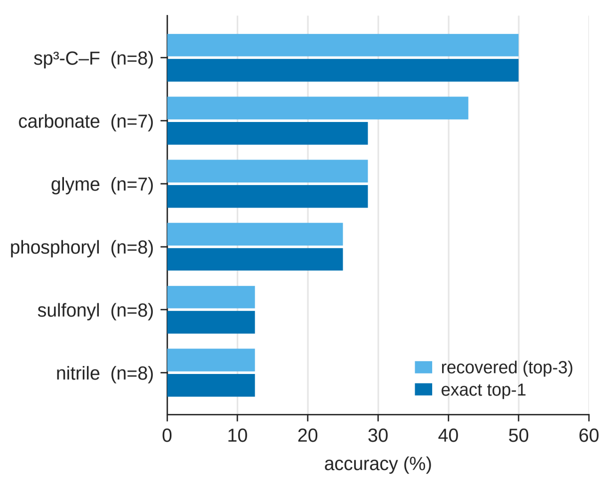

Fig. S4: IRSpectra-Bench-Electrolyte by battery-electrolyte class (n=46): sp^3-C–F easiest (50%), sulfonyl and nitrile hardest (12%).

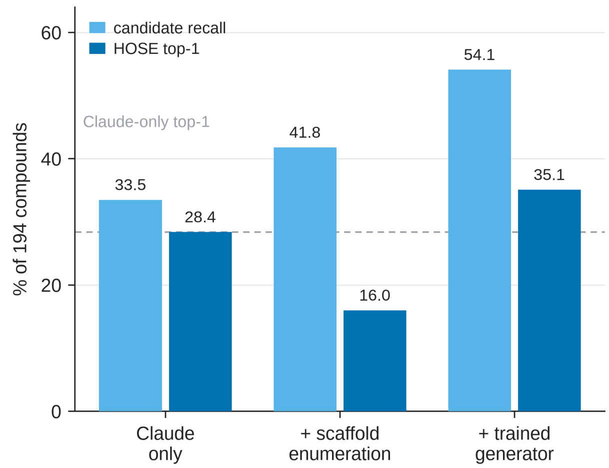

Fig. S5: Trained-generator probe (§5.6; a complement, not part of the training-free protocol). Candidate recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator: enumeration's near-degenerate isomers collapse the verifier (28.4→16.0%) while the generator's formula-correct candidates convert (28.4→35.1%).

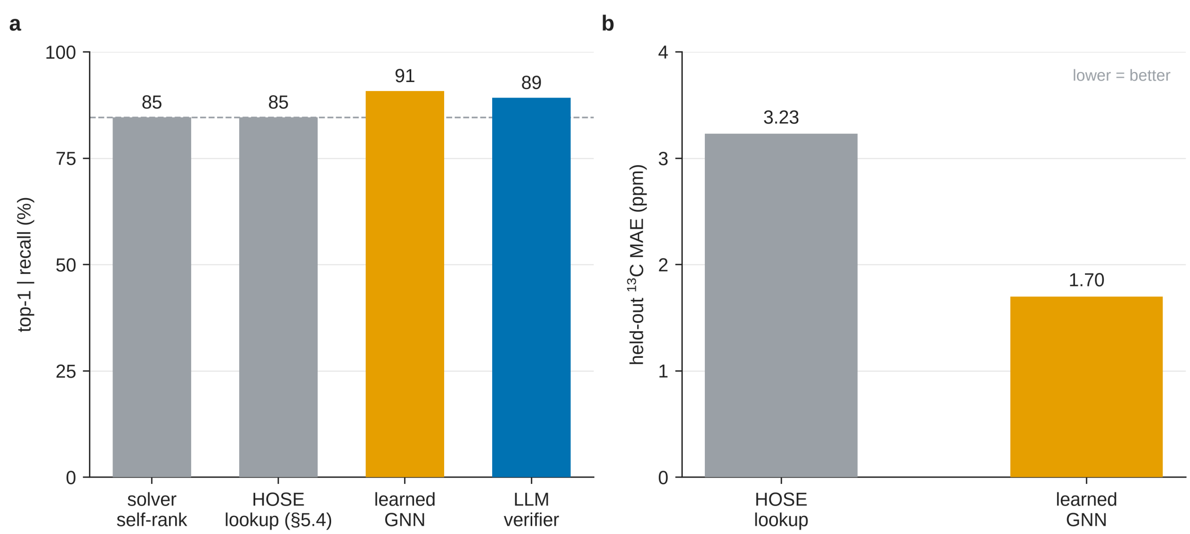

Fig. S6: Learned-verifier probe (§5.4; a complement, not part of the training-free protocol). (A) Conditional-on-recall top-1 (n=19) across four verifiers: a GNN trained on the same nmrshiftdb2 data as the HOSE lookup recovers the LLM verifier's 84% that the lookup (73%) could not. (B) Held-out ^{13}C MAE — the learned model is roughly 2× sharper (1.70 vs 3.23 ppm).